



ITI Cmos Analog IC Design
Lab 06
Differential Amplifier

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Section 7 online

1.2 Input Pair Sizing (Sizing Assistant)

3	L	350n
4	W	30.16u
7	VGS	940.7m

ID	20u	?
gm/ID	8/(0.909*0.6)	?
ro	10*(0.6/ID)	?
VDS	0.9	?
VS _B	0.3	?
Stack	1	?

W	30.16 u
L	350 n
gm/id	14.6
V*	136 mV
V _{TH}	870.2 mV
VGS	940 mV

the required CM input level :

$$V_{ICM} = V_{DD} - V_{GS_{3,4}} - V_{DS_1} = 1.8 - 0.94 - 0.3 = 0.56 V$$

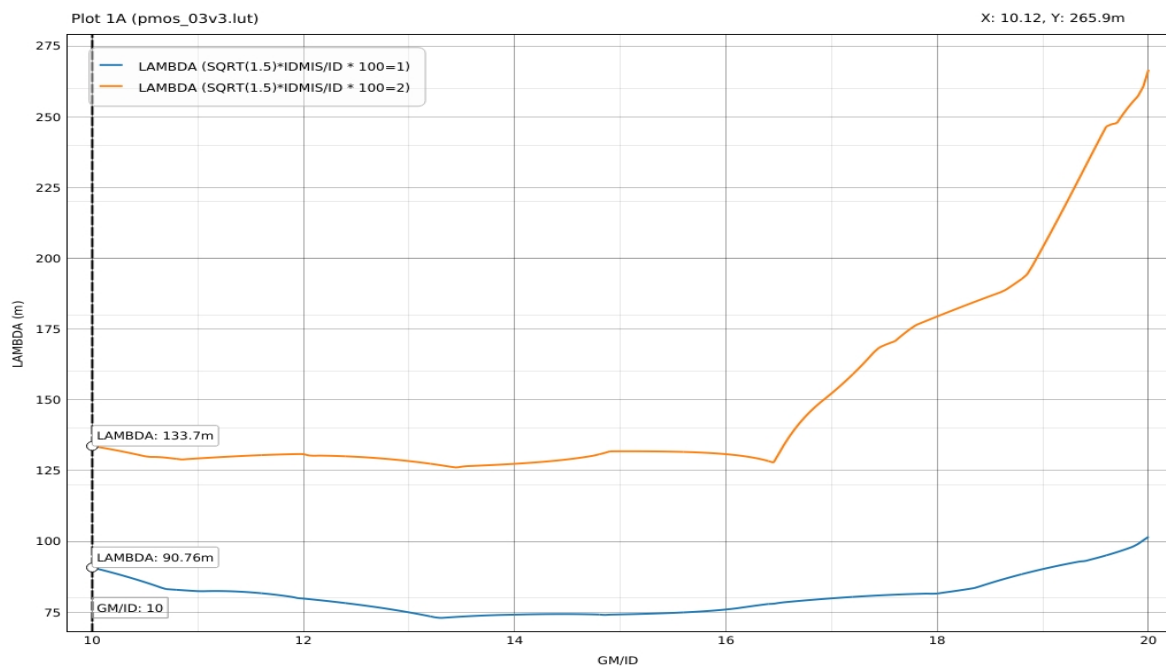
1.3 tail current mirror design

LUT	pmos_03v3	?
Corner	TT	☐ All ?
Temp (°C)	27.0	☐ All ?
Frequency	1	?
ID	20u	?
gm/ID	10:20	?
$\sqrt{1.5} \cdot \text{idmis}/\text{ID} \cdot 100$	1, 2	?
VDS	0.3	?
VSB	0	?
Stack	1	?

Sweep gm/ID vs. L, W, Area, and λ at constant $\sigma(\text{lout})/\text{lout}$ (1% and 2% curves).

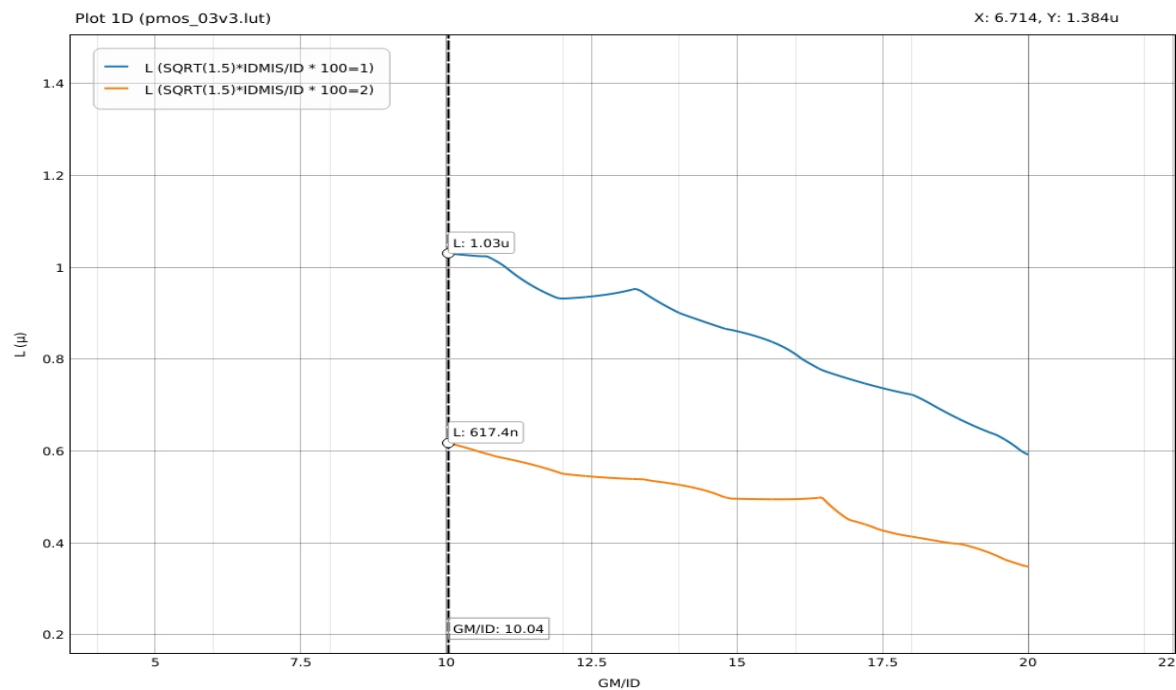
● λ_{bmbda} vs gm/id

$\sqrt{3}$, Corner=TT, Temp=27.0, freq=1, ID=20u, GM/ID=10:20, $\text{SQRT}(1.5) \cdot \text{IDMIS}/\text{ID} \cdot 100 = 1/2$, VDS=300m, V



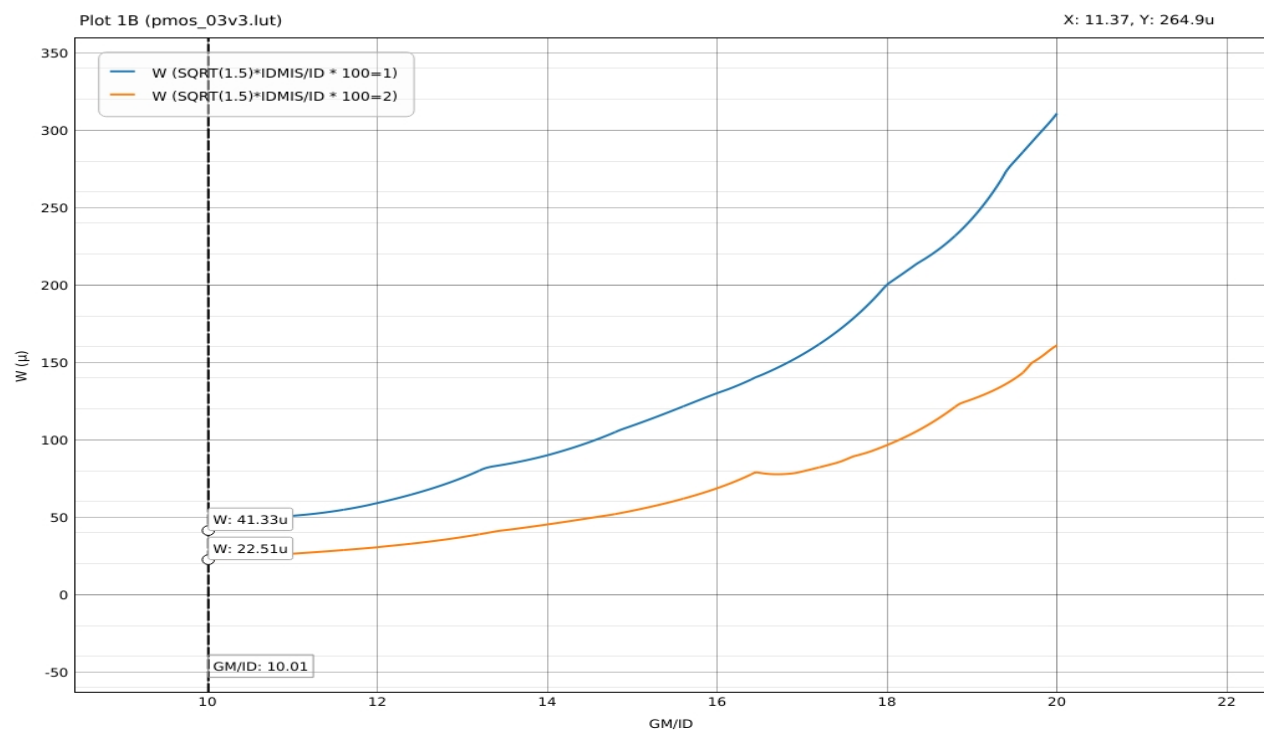
● L vs gm/id

, Corner=TT, Temp=27.0, freq=1, ID=20u, GM/ID=10:20, $\text{SQRT}(1.5) \cdot \text{IDMIS}/\text{ID} \cdot 100 = 1/2$, VDS=300m,



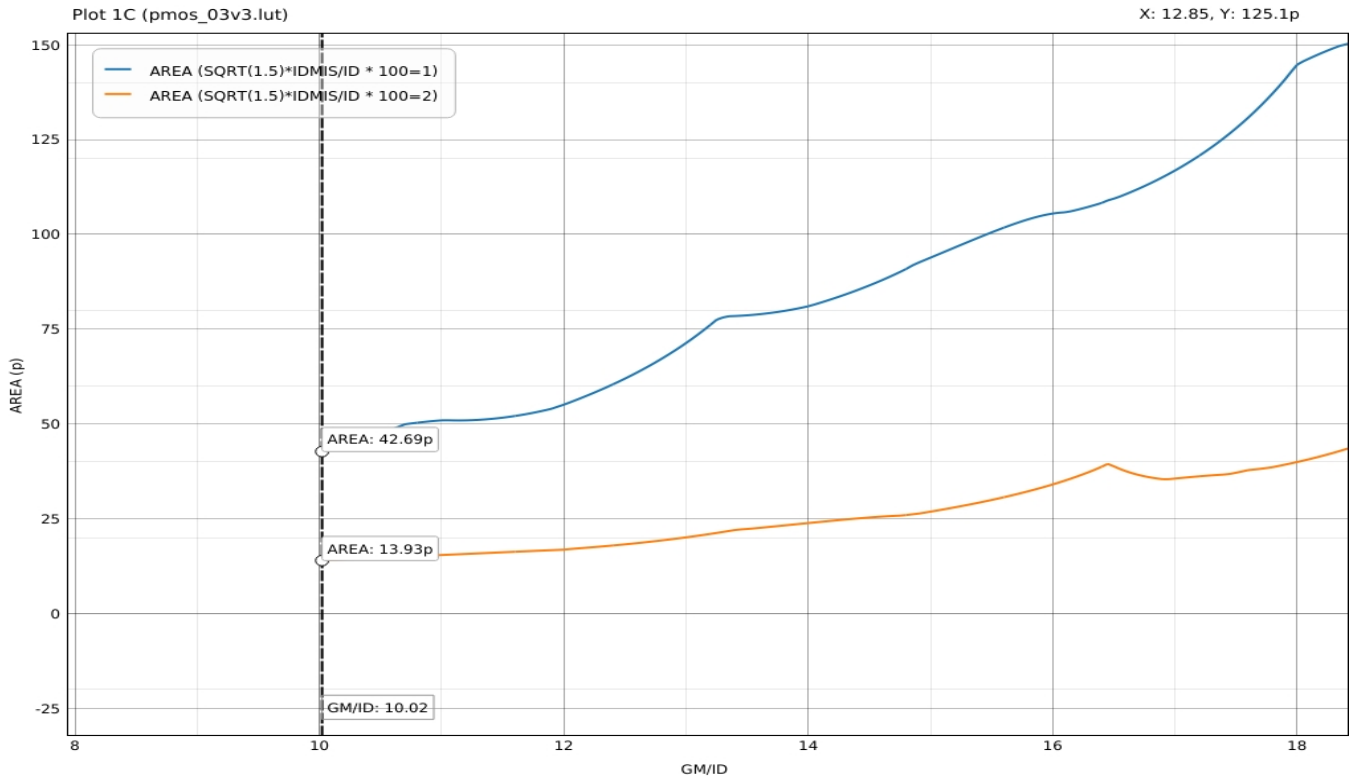
● W vs gm/id

/3, Corner=TT, Temp=27.0, freq=1, ID=20u, GM/ID=10:20, $\text{SQRT}(1.5) \cdot \text{IDMIS}/\text{ID} \cdot 100 = 1/2$, VDS=300m, V



● AREA vs gm/id

/3, Corner=TT, Temp=27.0, freq=1, ID=20u, GM/ID=10:20, SQRT(1.5)*IDMIS/ID * 100=1/2, VDS=300m, V



- We will choose the 1% curve (blue curve) to satisfy both mismatch lambda requirements.
- Compliance voltage is 200 mV so gm/id = 10

W	41.2 u
L	1 u
gm/id	10
V*	200 mV
Lambda	90 mV

1.10 CM Input Range

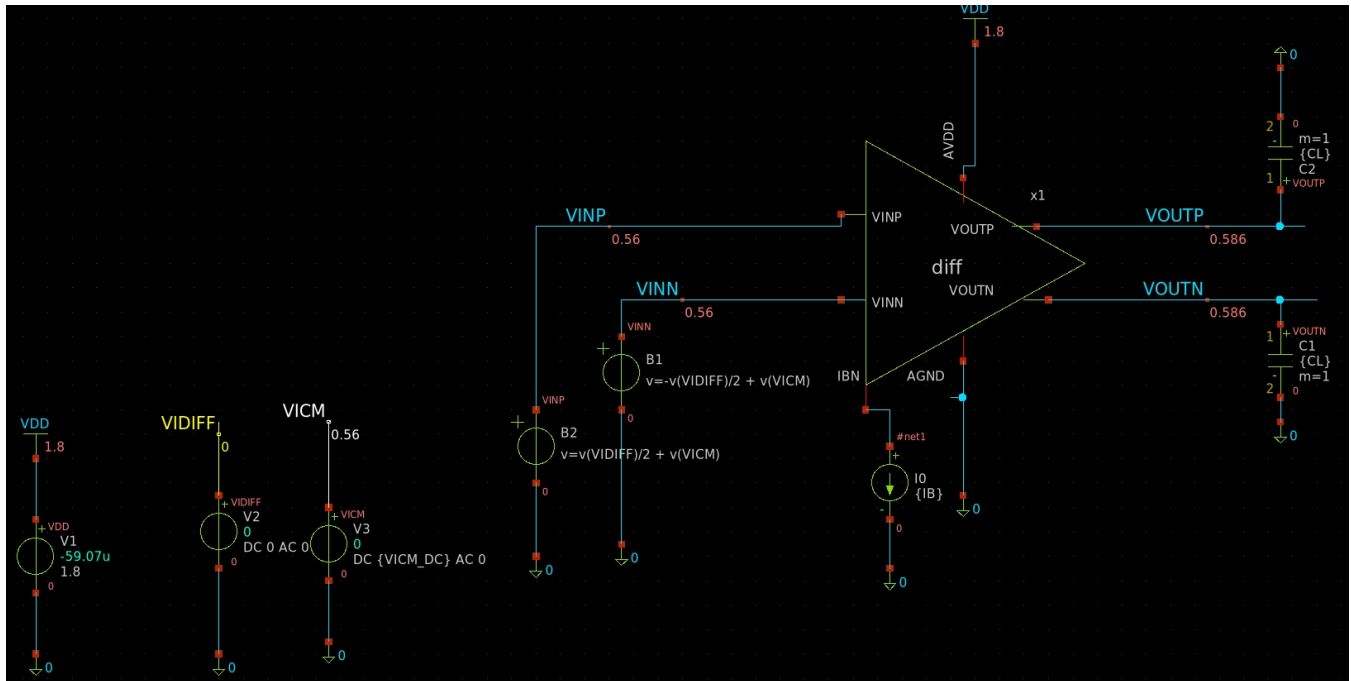
$$V_{RD} - V_{th_{3,4}} \leq V_{ICM} \leq V_{DD} - V_{GS_{3,4}} - V^*_1$$

$$-0.27 \text{ V} \leq V_{ICM} \leq .71 \text{ V}$$

Selected VICM = .56 V is in range

Part 2 — Differential Amplifier Simulation

2.1 Testbench



2.2 Operating Point (OP) Simulation

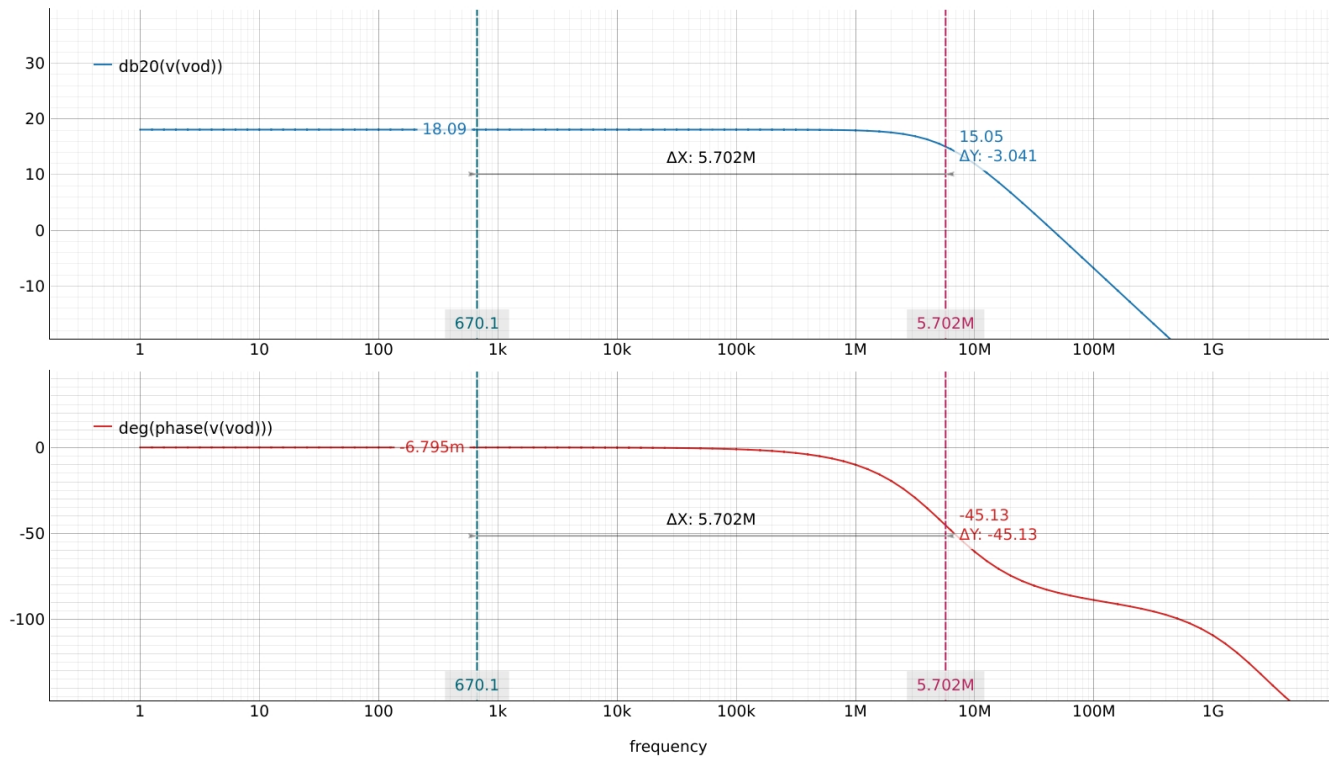
```
--- Model Operating Points ---
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm2.m0      m.x1.xm1.m0      m.x1.xm4.m0
model       pfet_03v3.13     pfet_03v3.13     pfet_03v3.12
id          2e-05            3.90694e-05      1.95347e-05
vgs         0.939506         0.939506         0.939965
vds         0.939505         0.300033         0.913923
vth         0.783441         0.783443         0.87412
vdsat       0.15285          0.152849         0.118625
gm          0.000203428       0.000396276     0.000286251
gds         4.47788e-07       3.44288e-06      3.13482e-06
gmbs        9.62184e-05       0.000187453     9.31482e-05

BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m.x1.xm3.m0
model       pfet_03v3.12
id          1.95347e-05
vgs         0.939965
vds         0.913923
vth         0.87412
vdsat       0.118625
gm          0.000286251
gds         3.13482e-06
gmbs        9.31482e-05

=== Bias Points ===
M1: gm/id = 10.1429 S/A, Vstar = 0.197183 V
M2: gm/id = 10.1714 S/A, Vstar = 0.19663 V
M3: gm/id = 14.6535 S/A, Vstar = 0.136486 V
M4: gm/id = 14.6535 S/A, Vstar = 0.136486 V
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
```

From the op result , all transistors operate in saturation since all vds is greater than V_{ds_sat}

2.3 Differential Small-Signal AC Characteristics



```
av_diff_db      = 1.80876e+01
av_diff_mag     = 8.02380e+00
bw_diff         = 5.68065e+06
```

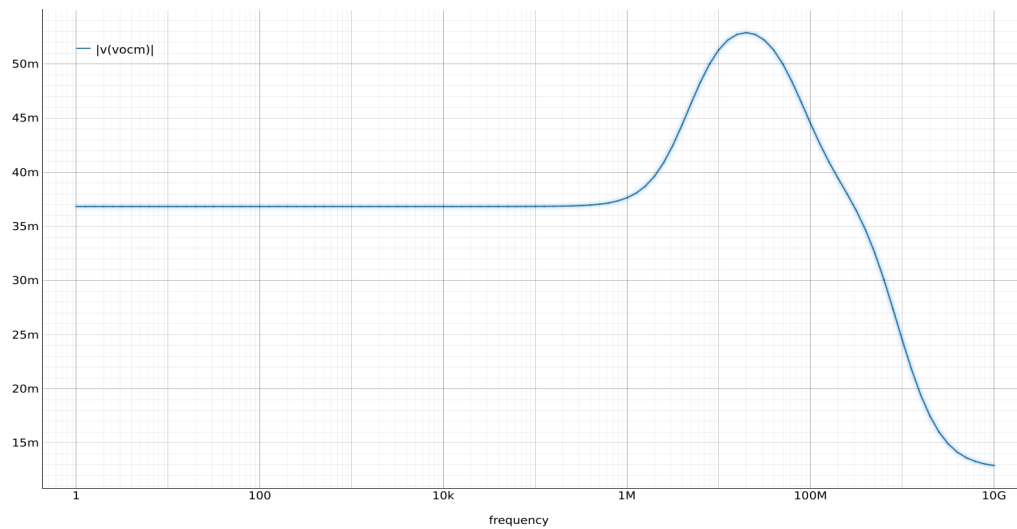
➤ Hand analysis

$$AV = gm(R_d) = 8$$

$$BW = 1 / 2\pi \cdot c_l \cdot R_D = 1 / 2\pi \cdot 1p \cdot 30k = 5.4 \text{ MHz}$$

Metric	Hand Analysis	Simulation	Error (%)
DC DIFF Gain, A_{vd} (dB)	8 (18 dB)	8 (18 dB)	0
BW_diff,	5.4 MHz	5.6 Mhz	3.5 %

2.4 CM Small-Signal



```

av_cm_db      = -2.86736e+01
av_cm_mag     = 3.68399e-02
bw_cm        = 8.87869e+08

```

➤ Hand analysis

$$AV = \frac{g_m R_D}{1 + 2(g_m + g_{mb}) R_{SS}} = \frac{286\mu \cdot 30k}{1 + 2(380\mu) 290k} = 38.68 \text{ m}$$

Metric	Hand Analysis	Simulation	Error (%)
DC CM Gain, Avcm (V/V)	38.68 m (-28.25 dB)	36.83 m (-28.6 dB)	5%

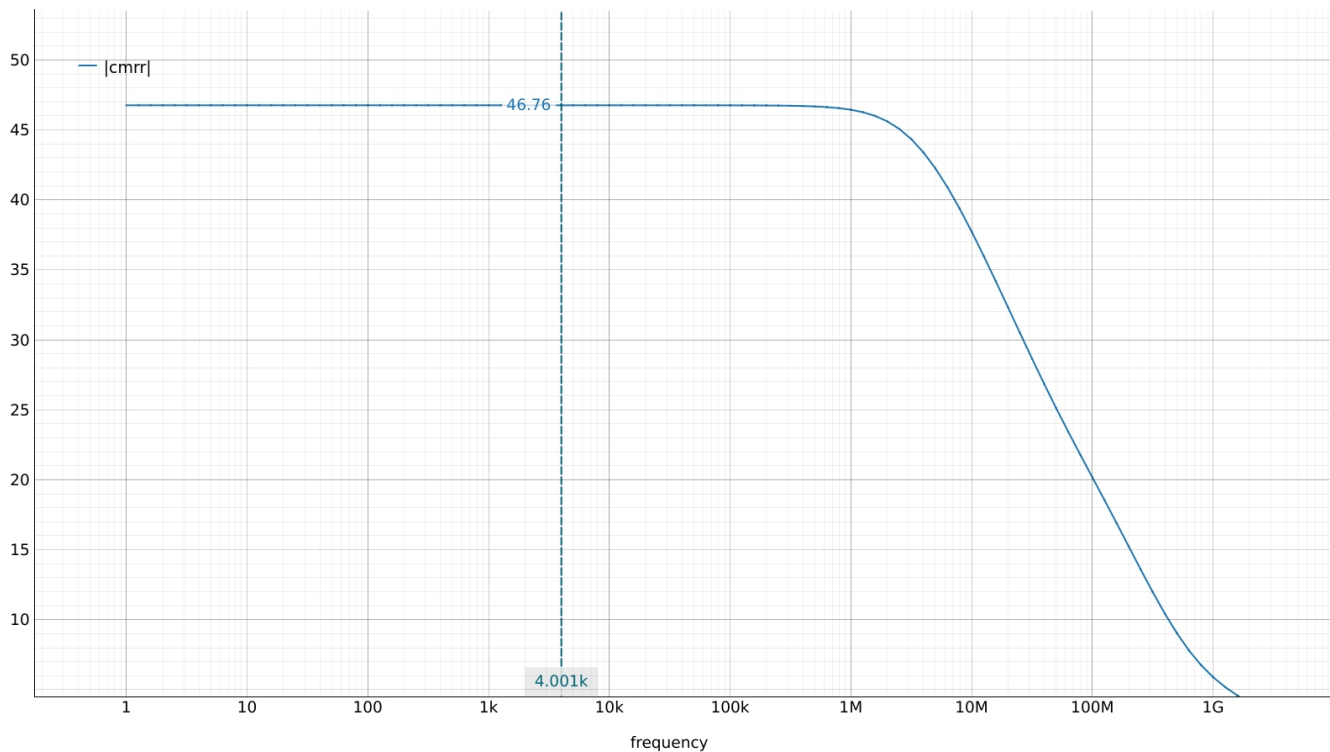
Is |Avcm| smaller than 1? why?

Yes , cause the resistance of tail current is so large , and when input try to increase together , tail will prevent them and the drain current will be aproximatly constant , so out put will be small.

Justify the variation of Avcm with frequency ?

at low frequency avcm is low because the tail resistance will reject all common mode signal
But at high frequency there will be some parasitic capacitanes will increase the AVCM again .

CMRR



➤ Hand analysis

$$\text{CMRR} = 2(g_m + g_{mb})R_{SS} = 2(380\mu)290k = 220.16 \text{ (46.76 dB)}$$

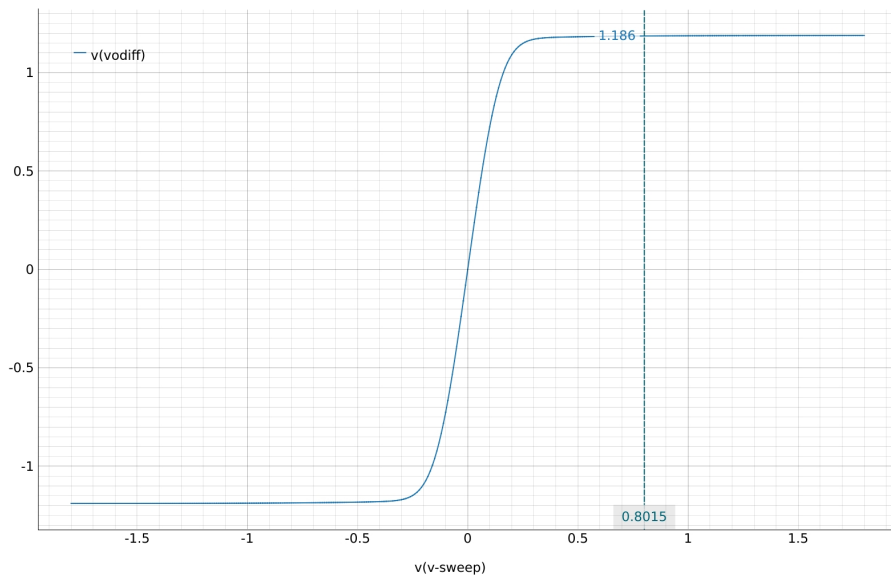
Metric	Hand Analysis	Simulation	Error (%)
CMRR @ DC (dB)	220 (46.75 dB)	220 (46.76 dB)	0

Justify the variation of CMRR with frequency?

at low frequency CMRR is high because A_{Vd} is very high and a_{vcm} low

at high frequency CMRR decreasing because of A_{Vd} is decreasing, and common mode try to increase due to parasitic cap so over all CMRR will decrease.

2.5 Differential Large-Signal Characteristics



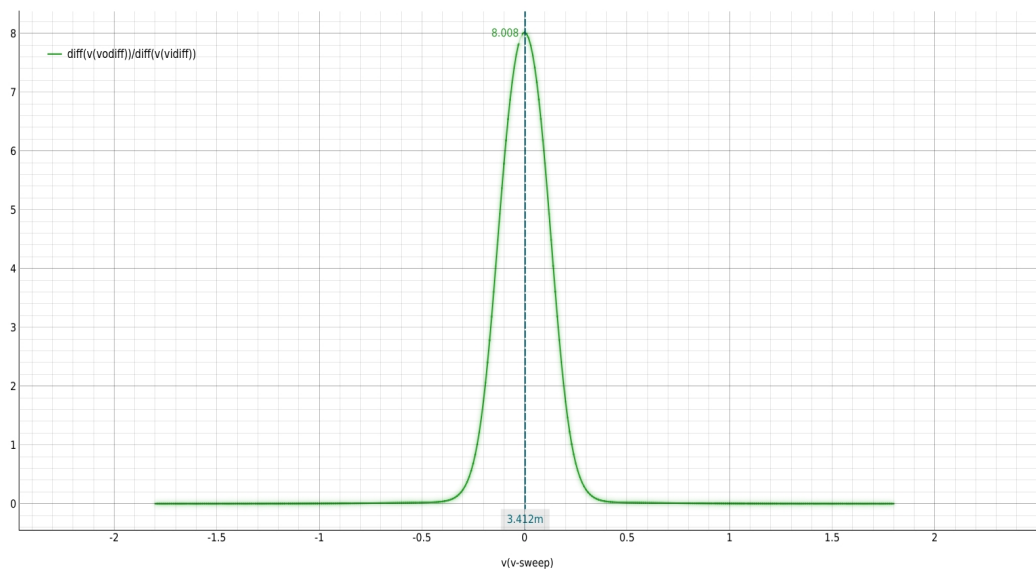
➤ Hand analysis

$$VODIFF_Max = 2 * ISS * RSS = 1.2 \text{ V}$$

$$VODIFF_min = -2 * ISS * RSS = -1.2 \text{ V}$$

Extreme Value	Hand Analysis	Simulation	Error (%)
Max VODIFF (V)	1.2	1.88	5.6%
Min VODIFF (V)	-1.2	-1.188	5.6%

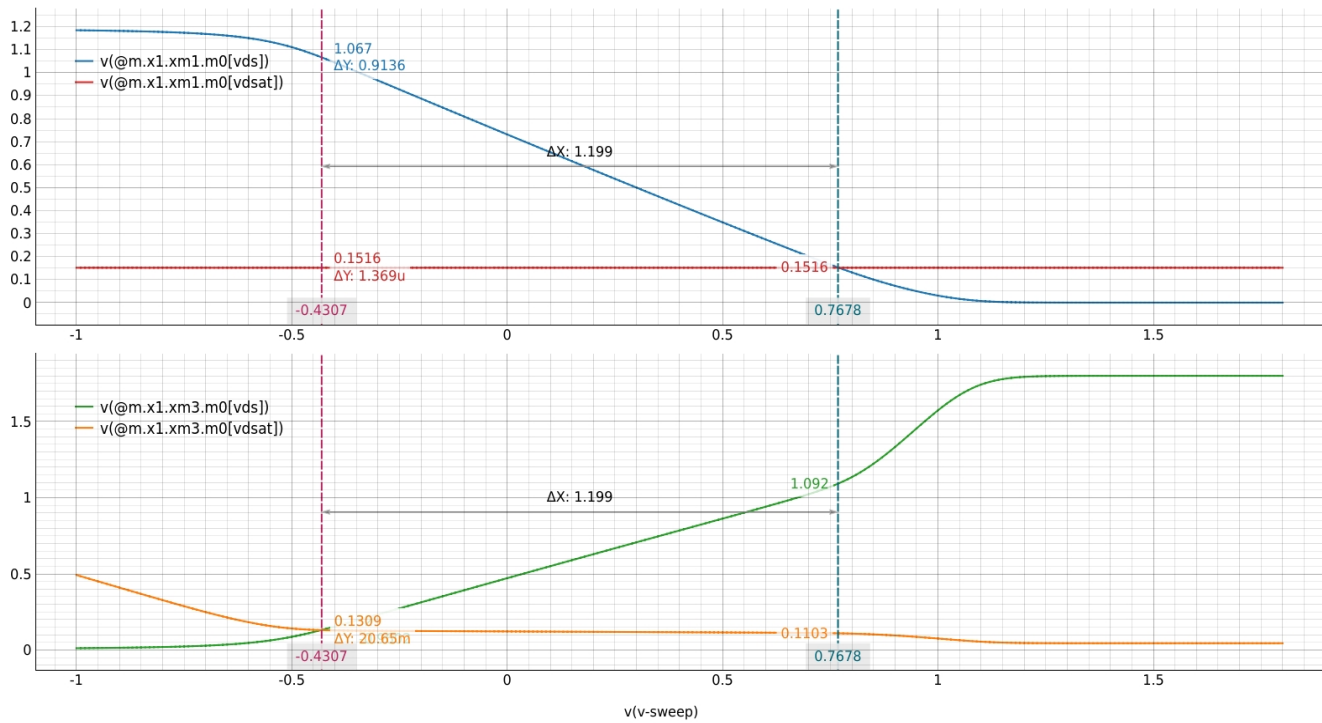
● deriv() of VODIFF vs VIDIFF



What is the peak value? Why ?

Peak value is 8 which is equal to the dc gain when $vid = 0$ that $equal = dv_{outd} / dv_{ind} = AV_d$

2.6 Common-Mode Large-Signal Characteristics — Region vs VICM



➤ Hand analysis

$$\text{VICM}_{\min} = V_{\text{RD}} - V_{\text{th}_{3,4}} = -0.27 \text{ V}$$

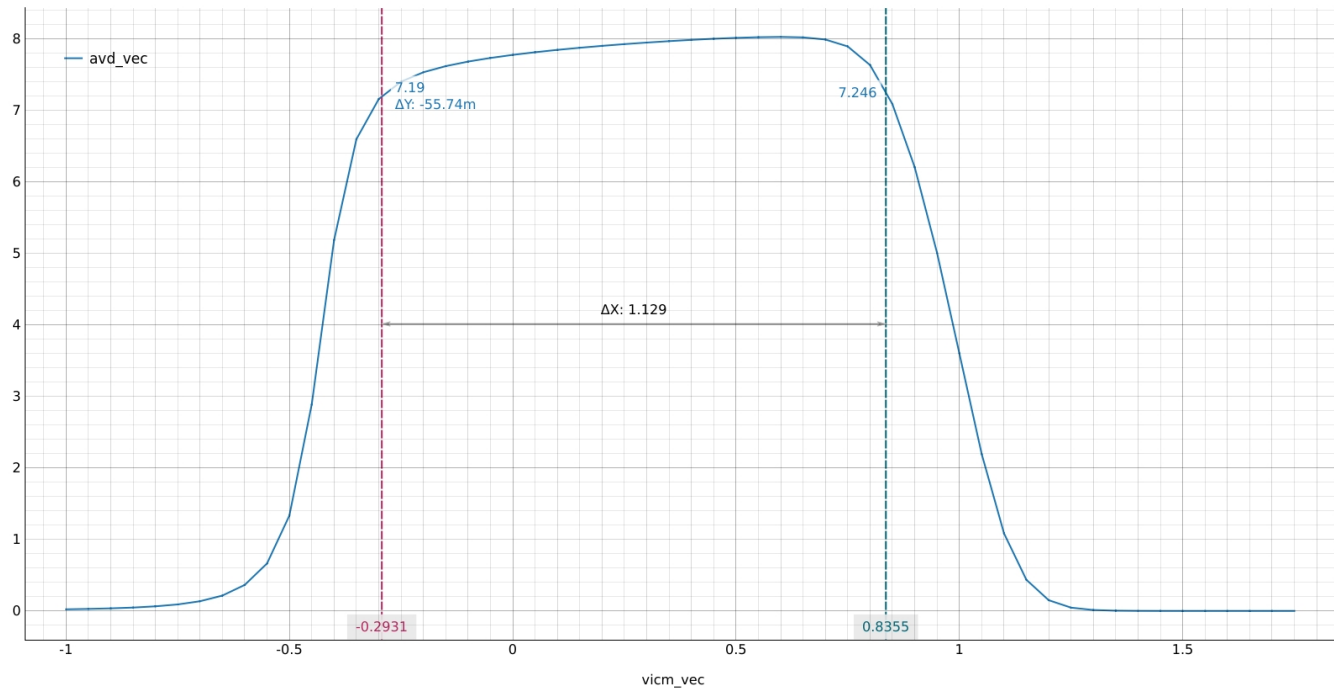
$$\text{VICM}_{\max} = V_{\text{DD}} - V_{\text{GS}_{3,4}} - V^*_1 = .71 \text{ V}$$

CM Input Range (CMIR)	Hand Analysis	Simulation	Error (%)
VICM,min (V)	-0.27	-0.4307	60%
VICM,max (V)	0.71	0.76	7%

Why does the minimum VICM not match the simple hand-analysis expectation?

I assumed that the v_{th} is constant in hand analysis, however v_{th} is not constant and varies with v_{sb} due to body effect, so the simulated $\text{vicm}_{\min}$ will be much lower than the hand analysis.

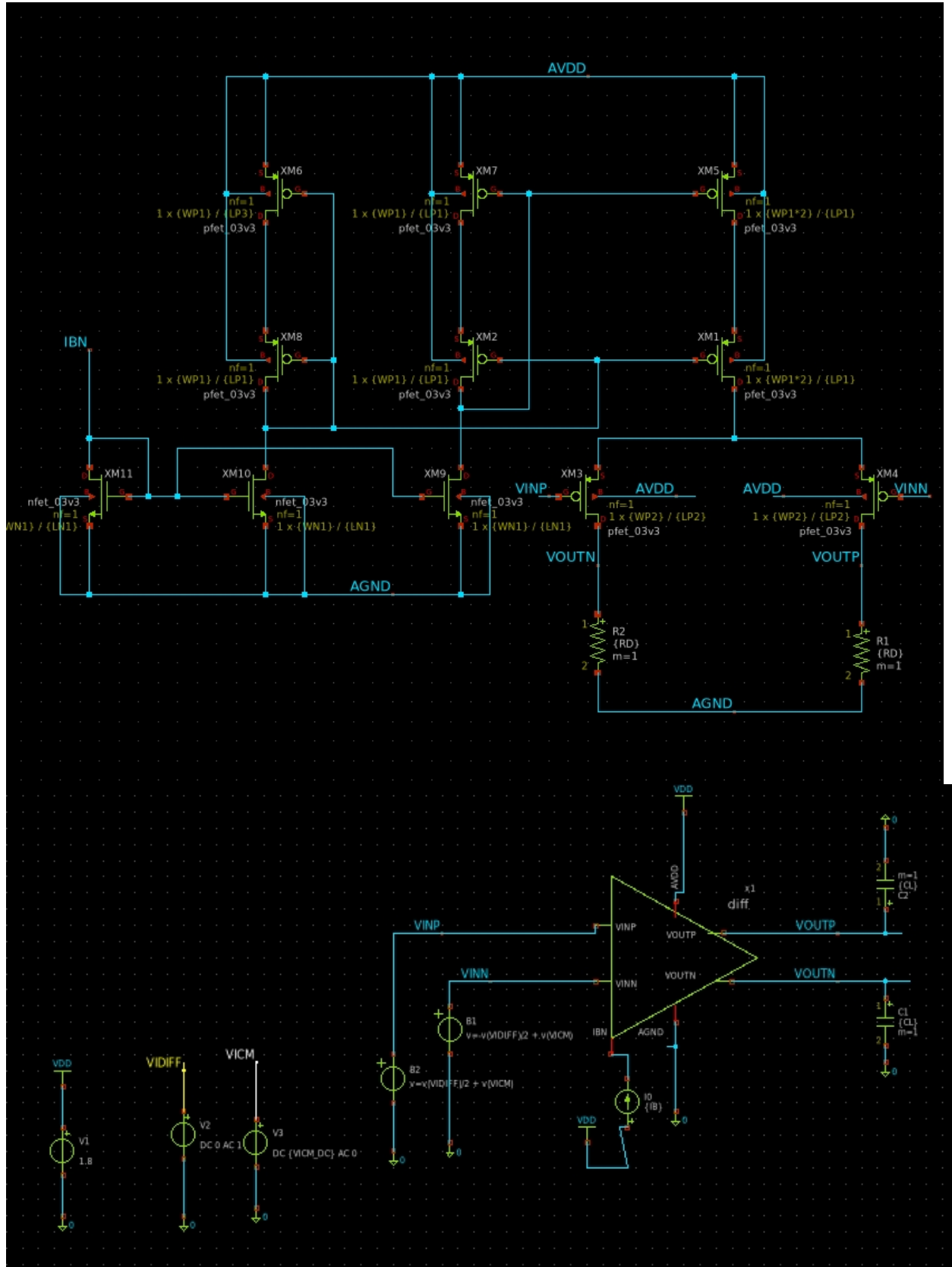
2.7 Common-Mode Large-Signal Characteristics — GBW vs VICM



CM Input Range (CMIR)	Hand Analysis	Simulation	Error (%)
VICM,min (V) — Vdsat method	-.27	-0.437	60%
VICM,max (V) — Vdsat method	0.71	.76	7 %
VICM,min (V) — Avd(90%) method	-.27	-0.2931	8.5%
VICM,max (V) — Avd(90%) method	0.71	0.8355	17 %

The 90% Avd method is much better than the previous method It show more aggrement with hand analysis .

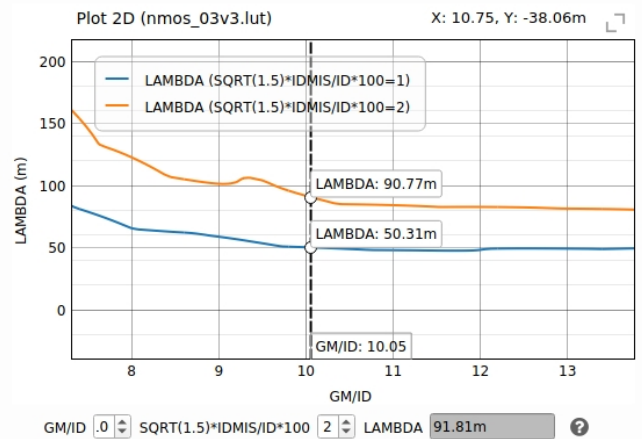
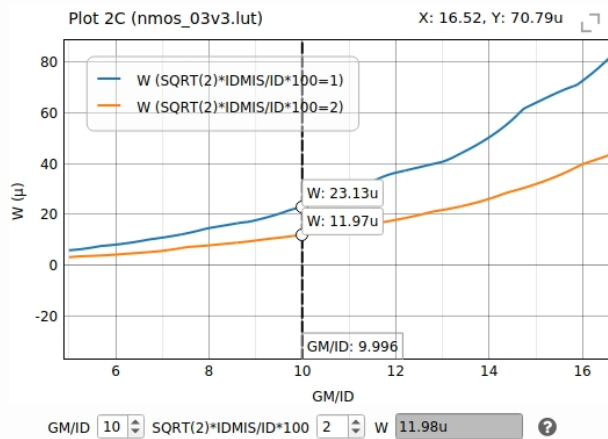
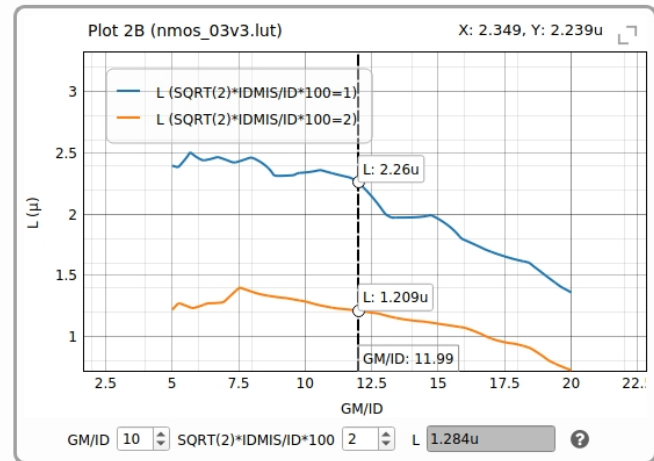
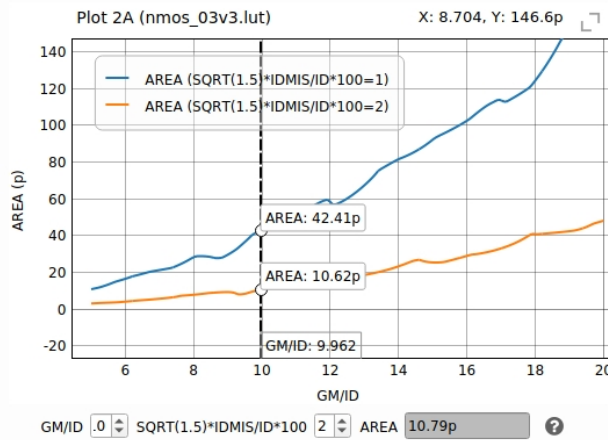
Bonus — cascode current mirror



➤ Nmos Current Mirror

To design nmos current mirror sweep gm/id with id= 20 um

LUT: ?
 Corner: All ?
 Temp (°C): All ?
 Frequency: ?
 ID: ?
 gm/ID: ?
 sqrt(2)*idmis/ID*100: ?
 VDS: ?
 VSB: ?
 Stack: ?



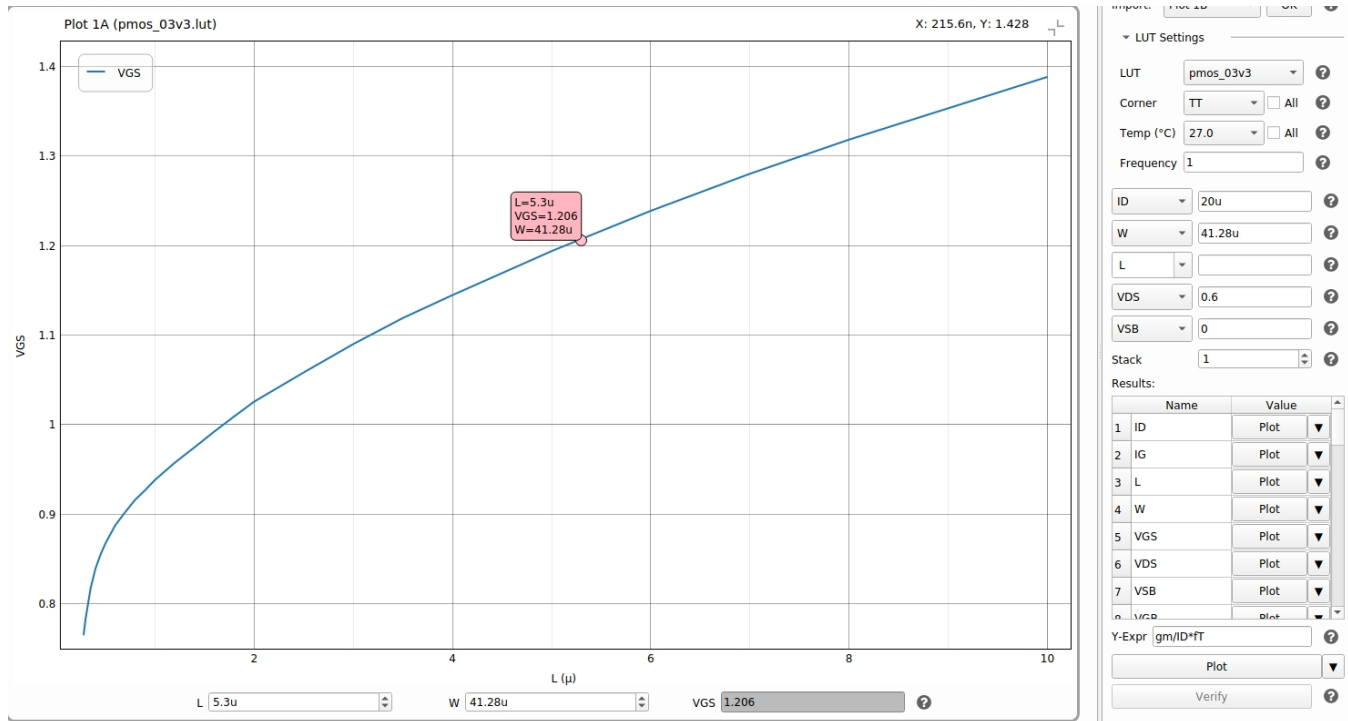
Minimizing area so taking 2% graph and gm/id = 10

W = 11.97u , and L = 1.209u

➤ **Magic battery**

$$VB = v_{gs1,2} + v^* + 50m = 1 + 200m = 1.2$$

keeping same Width and sweep on L to get $VB = 1.2$



So L for M6 will BE 5.3 u

➤ Op analysis

=== MOSFET Operating Points ===

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm11.m0	m.x1.xm9.m0	m.x1.xm10.m0
model	nfet_03v3.14	nfet_03v3.14	nfet_03v3.14
id	2e-05	2.00088e-05	1.98521e-05
vgs	0.841698	0.841698	0.841698
vds	0.841695	0.858474	0.582141
vth	0.682823	0.682823	0.682823
vdsat	0.178818	0.178818	0.178818
gm	0.000191789	0.000191849	0.000190705
gds	5.27549e-07	5.23798e-07	6.30534e-07
gmbs	7.42371e-05	7.42615e-05	7.38035e-05

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm6.m0	m.x1.xm8.m0	m.x1.xm7.m0
model	pfet_03v3.14	pfet_03v3.13	pfet_03v3.13
id	1.98521e-05	1.98521e-05	2.00088e-05
vgs	1.21785	1.02335	0.94152
vds	0.194498	1.02335	0.193512
vth	0.770006	0.872434	0.783942
vdsat	0.329977	0.15536	0.154428
gm	6.10106e-05	0.000205576	0.000196929
gds	5.86527e-05	4.28962e-07	9.86773e-06
gmbs	3.13791e-05	8.77434e-05	9.3236e-05

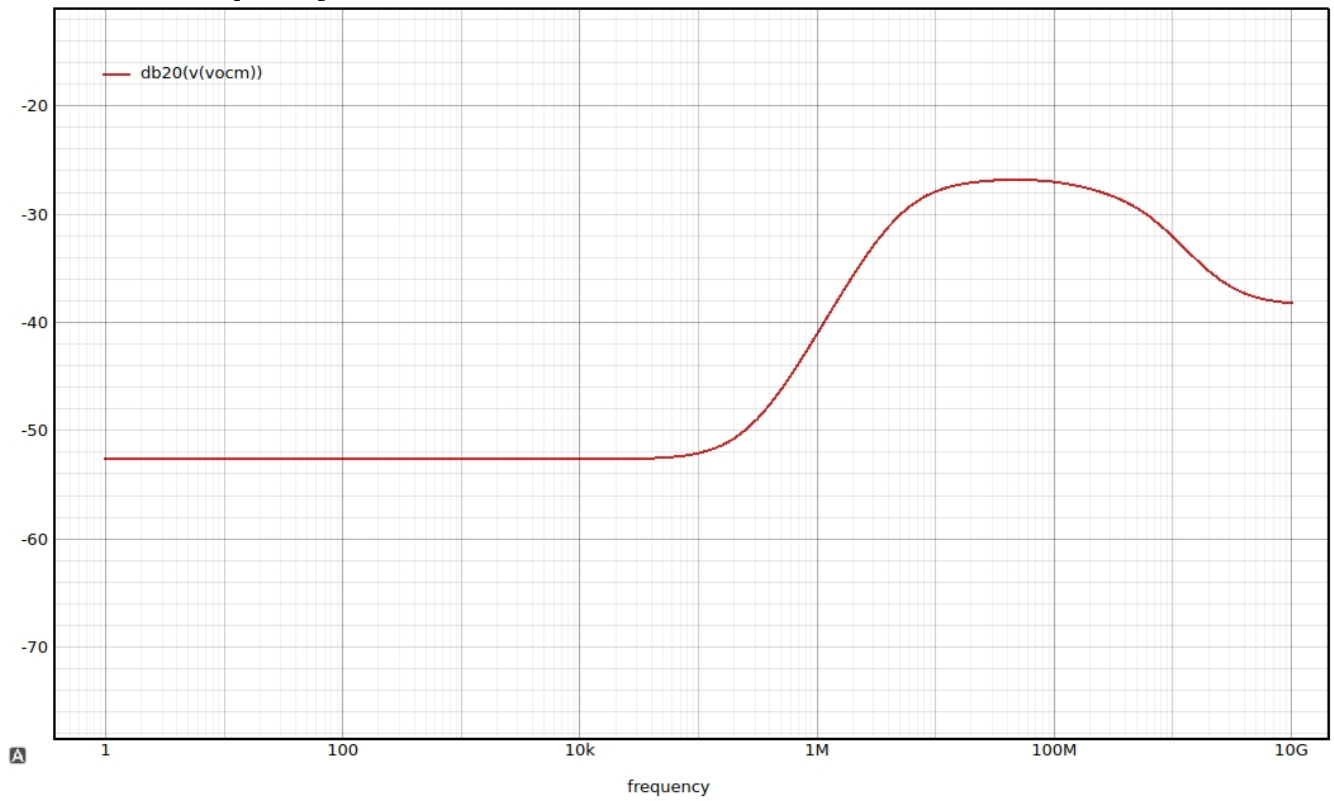
BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm5.m0	m.x1.xm2.m0	m.x1.xm1.m0
model	pfet_03v3.13	pfet_03v3.13	pfet_03v3.13
id	3.99757e-05	2.00088e-05	3.99757e-05
vgs	0.94152	1.02434	1.02594
vds	0.191913	0.748006	0.257285
vth	0.783942	0.87201	0.871321
vdsat	0.154428	0.156321	0.157885
gm	0.000393014	0.000206014	0.000404899
gds	2.05087e-05	5.48068e-07	6.02702e-06
gmbs	0.000186089	8.79689e-05	0.000173078

BSIM4v5: Berkeley Short Channel IGFET Model-4

device	m.x1.xm4.m0	m.x1.xm3.m0
model	pfet_03v3.12	pfet_03v3.12
id	1.99879e-05	1.99879e-05
vgs	0.990799	0.990799
vds	0.751162	0.751162
vth	0.921657	0.921657
vdsat	0.12247	0.12247
gm	0.000289068	0.000289068
gds	3.51143e-06	3.51143e-06
gmbs	8.76346e-05	8.76346e-05

➤ **AV cm Vs frequency**



Av CM: -52.6486 dB, BW: Hz
CMRR: 70.5409 dB